

Ruaridh Murdoch

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Predicted first-class MEng Design Engineering student at Imperial College London, focused on mechatronics and human-centred design. I'm drawn to where hardware meets product: a design decision has to work mechanically, electrically, and for the person using it. I move easily between CAD, rapid prototyping and embedded systems, and I care about manufacturability as much as the first working prototype. I keep going until it does what I set out to do. Looking for a placement where I can contribute to real products from concept through to build.

Education

Imperial College London

MEng Design Engineering, Dyson School of Design Engineering

London, UK

Oct 2024 — Jun 2028

- Year 2: 77.9% - Year 1: 72.7%
- Relevant coursework: Physical Computing, Electronics: Signals & Systems, Solid Mechanics, Finite Element Analysis, Thermofluids, Product Design Engineering, Human-Centred Design Engineering, Computing: Applications

Projects

SwivelCut: Motorised Cutting Arm

2026

- Identified through cross-school research (11 interviews, 92-person survey) that students' cardboard prototypes consistently fell short of their designs - manual tools lacked the accuracy to cut what they had drawn.
- Built a 2-joint motorised arm with swappable cutting and tracing heads, sized and tested to cut cleanly across a full A3 sheet; a sprung blade mount eliminated drag spikes from cardboard thickness variation, reducing peak cutting force from 28 N to 8 N.
- Delivered a £299 retail-costed, fully compliant product, validated through user testing with 18 Design Technology students across Years 7 - 12.

Teach-by-Touch Robotic Arm

2025

- Built a 4-joint robotic arm that learns motion by being physically moved - record a sequence by hand, insert a cartridge, press play to repeat it; developed under a low-cost constraint using servo-equivalent joints constructed from DC motors, 3D-printed gear reductions and potentiometer feedback.
- Diagnosed and resolved mechanical failures - shoulder stall fixed by doubling the gear reduction, gripper locking fixed by adding bearings.
- Programmed a four-state finite state machine with time-synchronised playback so motion replays at original speed; sequences stored to SD card and recalled via swappable colour-identified cartridges.

Prosthetic Running Blade

2026

- Designed and analysed a CFRP sprinting blade against three criteria (mass, fatigue, frequency); the initial design failed all three, with stress at 2,924 MPa against a 600 MPa limit.
- Used beam theory to predict the required geometry changes before re-simulating, cutting stress to 419 MPa and achieving a ~2x frequency increase, predicted to within 1.5% of the FEA result.

Aerodynamic Cabriolet: CFD

2026

- Designed a four-seat open-top cabriolet around rear wake management - the dominant drag source in roof-down driving - and iterated the geometry in Ansys CFD, reducing drag from 758 N to 653 N (-14%) and Cd from 0.458 to 0.405.
- Validated CFD against a 1:23 scale wind tunnel test; Cd predicted at 0.458 against a measured 0.461, and a lift discrepancy was traced to the test method rather than the design.

Skills

- **Electronics & embedded:** Arduino (C++), MicroPython, ESP32, circuit prototyping, sensor integration, DC / stepper / servo motor control, PID and finite-state-machine control
- **Mechanical & CAD:** Fusion 360, tolerance & fit, gear & mechanism design, DFM / DFA, 3D printing, rapid prototyping, technical drawing
- **Simulation & analysis:** FEA and CFD in Ansys (Mechanical, Fluent), MATLAB, hand-calculation validation
- **Software:** Python, C#, C++, JavaScript, Git, Unity
- **Design & research:** user research & interviews, survey design, journey & stakeholder mapping, benchmarking, lifecycle & materials analysis, Figma, UI design

Experience

Undergraduate Teaching Assistant

Oct 2025 — Jul 2026

Imperial College London

London, UK

- Mentored over 90 first-year Design Engineering students each week across maths, mechanics, materials, computing and design coursework.

AI Model Trainer

Jul 2025 — Present

DataAnnotation

Remote

- Evaluate and annotate AI model outputs across reasoning, mathematics and programming, identifying failure modes and edge cases to improve model accuracy.

Volunteer Bicycle Mechanic

2023 — 2024

Recyke-a-Bike

Stirling, UK

- Diagnosed faults and carried out full mechanical repairs on donated bikes for community redistribution: drivetrains, brakes and structural components.

Coding Club Mentor

2022 — 2023

The Glasgow Academy

Glasgow, UK

- Led coding sessions for secondary students, breaking down programming concepts and guiding problem-solving.

Interests

- **Game jams:** won Duck Sauce Game Jam two years running (48-hour Unity and C# builds), shipping fully playable original games under time pressure.
- **Climbing & outdoors:** NIBAS / NICAS-certified lead climber, former Glasgow Youth Squad competitor, completed the Lakeland Mountain Marathon.
- **Sailing:** RYA Day Skipper, sailed Scottish coastal waters independently.